

Introduction

Motivation

Closing the reality gap is a crucial research direction in achieving true embodied intelligence. To attain this goal, multiple factors must be considered: in real-world scenarios, we encounter uncertainties stemming from sensor noise, modeling errors, and delays. Additionally, agents deployed on real robotic systems must operate without damaging their components, the environment, or causing harm to people.

The game of air hockey presents a complex and dynamic challenge where a robotic player must be highly reactive, adhere to constraints, and exhibit long-term planning capabilities. This scenario compels engineers to push the boundaries of robotic capabilities while ensuring safety.

With these challenges in mind, the Robot Air Hockey competition was organized. This competition serves as a platform for researchers in the field of robot learning to engage with a realistic robotic task. Teams participating in the competition design and build their air hockey agents, competing against each other in various subtasks (in simulation) and ultimately in full games (both in simulation and in the real world).

To be successful, an agent must solve multiple sub-problems, such as scoring goals against an opponent, effectively defending its own goal, and devising winning strategies. The inherent sub-problem composition of air hockey inspires this work, where a hierarchical approach to the game is investigated. Reinforcement learning techniques are applied to each sub-problem that contributes to a complete air hockey match. These specialized agents are then integrated into a unified high-level agent that continuously selects the optimal low-level agent to deploy during the match.

Goal

The goal of this work is to develop an agent capable of playing entire air hockey games while adhering to all operational constraints. Specifically, the agent must control a general-purpose manipulator, achieving the best possible performance in terms both of offense and defense. It is crucial that the agent maintains these performance levels without violating any constraints, ensuring that no drop in performance or damage concerns arise when deployed on a real robot. The agent must:

- react quickly and accurately to the puck's movements.
- Maintain strategic positioning and make decisions that balance offensive and defen-

sive play.

- Operate within safety margins, ensuring that the manipulator does not exceed its physical limits or damage constraints.

Thesis Structure

The thesis is structured as follows:

1. **Preliminary Background:** This section introduces the main theories of Reinforcement Learning and presents state-of-the-art Deep Reinforcement Learning algorithms used for continuous control, as well as a brief overview of hierarchical reinforcement learning frameworks.
2. **Related Works:** This chapter reviews existing literature on trajectory optimization algorithms used for robotics control and describes the main challenges in applying reinforcement learning techniques to robotics.
3. **Air Hockey Competition:** This chapter describes the competition, its rules, and the provided framework.
4. **Methodology:** This chapter details the design and implementation of the agents used during the competition, including the hierarchical approach and reinforcement learning techniques applied.
5. **Experimental Results:** This chapter presents the results obtained during the competition and discusses the improvements made post-competition.

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